Salesforce Email Notification System using Apex and Visualforce

Aim

Develop an Apex program in Salesforce that sends an email notification to a specified email address using Salesforce Messaging class with frontend using Visualforce Pages.

Features:

- Enter recipient email address
- Enter subject
- Enter message body
- Validate email address
- Display invalid email message
- Send email successfully
- Optional attachment support

Step 1: Create Salesforce Developer Account

Open Salesforce Developer Signup page: https://developer.salesforce.com/signup

Create your developer account and login.

Step 2: Open Developer Console

After login:

1. Click Setup (⚙️)
2. Select Developer Console

Developer Console opens in a new window.

Step 3: Enable Email Deliverability

Important Step: Without this step email will not send.

Go to:

Setup → Search → Deliverability

Set:

Access Level = All Email

Click Save.

Step 4: Create Apex Controller Class

Open:

Developer Console → File → New → Apex Class

Class Name:

EmailController

Paste the following code:

```
public class EmailController {

    public String toAddress {get; set;}
    public String subjectText {get; set;}
    public String bodyText {get; set;}

    // Send Email Method
    public void sendEmail() {

        // Validate Email
        if(toAddress == null || !Pattern.matches(
            '^[A-Za-z0-9+_.-]+@[A-Za-z0-9.-]+$',
            toAddress
        )) {

            ApexPages.addMessage(
                new ApexPages.Message(
                    ApexPages.Severity.ERROR,
                    'Invalid Email Address!'
                )
            );

            return;
        }
```

```
        // Create Email Object
        Messaging.SingleEmailMessage mail =
            new Messaging.SingleEmailMessage();

        // Recipient Email
        mail.setToAddresses(
            new List<String>{toAddress}
        );

        // Subject
        mail.setSubject(subjectText);

        // Message Body
        mail.setPlainTextBody(bodyText);

        // Send Email
        Messaging.sendEmail(
            new List<Messaging.SingleEmailMessage>{mail}
        );

        // Success Message
        ApexPages.addMessage(
            new ApexPages.Message(
                ApexPages.Severity.CONFIRM,
                'Email Sent Successfully!'
            )
        );
    }
}
```

Click File → Save.

Step 5: Create Visualforce Page

Open:

Developer Console → File → New → Visualforce Page

Page Name:

SendEmailPage

Paste the following code:

```
<apex:page controller="EmailController">

    <apex:form>

        <apex:pageBlock title="Email Notification System">

            <apex:pageMessages />

            <apex:pageBlockSection columns="1">

                <apex:outputLabel value="Recipient Email"/>

                <apex:inputText value="{!toAddress}" />

                <apex:outputLabel value="Subject"/>

                <apex:inputText value="{!subjectText}" />

                <apex:outputLabel value="Message Body"/>

                <apex:inputTextarea value="{!bodyText}" rows="5"/>

            </apex:pageBlockSection>

            <apex:commandButton
                value="Send Email"
                action="{!sendEmail}"
            />

        </apex:pageBlock>

    </apex:form>

</apex:page>
```

Click File → Save.

Step 6: Run the Visualforce Page

Go to:

Setup → Visualforce Pages

Open:

SendEmailPage

Click Preview.

Step 7: Test the Application

Enter the following values:

Field	Example
Recipient Email	abc@gmail.com
Subject	Test Email
Message Body	Hello from Salesforce

Click:

Send Email

Successful Output

Message displayed:

Email Sent Successfully!

The email will be received on entered email address.

Invalid Email Test

Enter invalid email:

abc@

Click Send Email.

Output:

Invalid Email Address!

Optional Attachment Support

If attachment support is required, add the following code before Messaging.sendEmail():

```
Messaging.EmailFileAttachment attach =
    new Messaging.EmailFileAttachment();

attach.setFileName('sample.txt');

attach.setBody(
    Blob.valueOf('This is attachment content')
);

mail.setFileAttachments(
    new Messaging.EmailFileAttachment[] {attach}
);
```

Important Salesforce Components Used

Apex Class

Used for backend logic and sending email.

Visualforce Page

Used for frontend user interface.

Messaging.SingleEmailMessage

Used to create and send email.

ApexPages.addMessage()

Used to display success and error messages.

Expected Output

The application should:

- Send email successfully
- Validate email address
- Show success/error messages
- Use Visualforce frontend
- Use Apex Messaging class

Viva Questions and Answers

1. What is Apex?

Apex is Salesforce backend programming language.

2. What is Visualforce?

Visualforce is Salesforce framework used to create custom pages.

3. What is Messaging.SingleEmailMessage?

It is a Salesforce class used for sending emails.

4. Why is email validation required?

To prevent sending emails to invalid addresses.

5. What does ApexPages.addMessage() do?

It displays success or error messages on Visualforce pages.

6. What is the use of setToAddresses()?

It specifies recipient email address.

7. What is the use of setSubject()?

It sets email subject.

8. What is the use of setPlainTextBody()?

It sets email message content.

Conclusion

Successfully developed a Salesforce Email Notification System using Apex and Visualforce Page with email validation and messaging functionality.